
Improving Confidence Calibration in Vision-Language Models via Post-hoc Uncertainty Estimation

Tingnan Bi

Department of Computer Science
Western University
tbi4@uwo.ca

Sara Rostamidaroukola

Department of Neuroscience
Western University
srostam7@uwo.ca

Nouran Sakr

Department of Computer Science
Western University
nsakr2@uwo.ca

Abstract

Vision-language models have recently achieved strong performance on multi-modal reasoning tasks such as visual question answering [1], yet their predictive confidence often remains poorly calibrated. In practice, these models may assign high confidence to incorrect answers, which limits their reliability in downstream applications. In this work, we evaluate the performance of a lightweight Laplace approximation to determine whether this method of post-hoc calibration can improve the confidence estimates of BLIP-2 on visual question answering. We then compare its performance on the VQA v2 dataset to two other post-hoc uncertainty estimation methods: temperature scaling and Monte Carlo (MC) dropout, and further study confidence-based abstention using a fixed abstention threshold. Our experiments show that post-hoc calibration can substantially reduce confidence error without modifying the model’s pretrained backbone. Among the methods we evaluate, temperature scaling provides the strongest improvement in calibration metrics, achieving the lowest expected calibration error and negative log-likelihood. MC Dropout yields the highest answer accuracy and also improves calibration relative to the uncalibrated baseline, although its calibration performance is weaker than temperature scaling in our final setting. By contrast, the lightweight Laplace approximation offers only limited benefit, suggesting that simple local posterior approximations may be insufficient for this multimodal setting.

1 Introduction

Visual question answering (VQA) tasks require a model to answer natural-language questions about an image, combining visual understanding with linguistic reasoning [1]. Vision-language models (VLMs) have become the standard approach for such multimodal problems, jointly processing images and text for tasks like VQA, image captioning, and open-ended multimodal dialogue [2]. To produce a reliable answer, a VLM should be well-calibrated, which means that it reports high confidence only when visual and textual evidence genuinely supports its answer, so that a predicted probability of 0.9, for example, corresponds to an empirical correctness rate of about 90% [3].

In practice, however, modern neural networks are often overconfident, meaning that they produce high predicted probability (confidence) even when the empirical correctness rate (accuracy) does not support that result [3]. In VQA, inherently ambiguous questions, multiple plausible answers, and

dataset biases can all lead models to assign high probability to responses that are only weakly supported by real-world likelihood [1]. Thus, finding methods of confidence calibration, or matching the model’s estimated probabilities with real-world likelihoods, becomes critical for multiple reasons. Firstly, in downstream decision-making pipelines, confidence is often used to prioritize predictions or trigger fallback mechanisms. Secondly, reliable uncertainty estimates are required for selective prediction, where the model should abstain when it is not sufficiently certain. Finally, in open-ended VQA, models may generate plausible-sounding but incorrect answers, which makes overconfidence especially misleading.

In this paper, our goal is two-fold: to improve scalar calibration metrics, such as expected calibration error (ECE) and negative log-likelihood (NLL), and to enable more reliable abstention behavior in uncertain cases. We aim to evaluate lightweight Laplace approximation to determine whether this post-hoc calibration method can improve the confidence estimates of a frozen BLIP-2 model on visual question answering. We compare the results to two other post-hoc calibration approaches: temperature scaling and MC Dropout. We cast confidence calibration for VLM-based VQA as a post-hoc uncertainty estimation problem over candidate answers, and evaluate all three methods within the same BLIP-2 pipeline. Our results show clear trade-offs across methods: temperature scaling offers the strongest calibration improvement at minimal cost, MC Dropout achieves the highest answer accuracy while improving calibration relative to the baseline, and the lightweight Laplace approximation provides only limited benefit. These findings suggest that, while a local posterior approximations may be insufficient for our multimodal setting, post-hoc calibration is a practical way to improve the reliability of pretrained VLMs without retraining the backbone.

2 Related Work

Confidence calibration has been widely studied in supervised learning. Modern neural networks are often miscalibrated despite strong predictive accuracy, and temperature scaling as a simple and effective post-hoc calibration method [3]. A complementary line of work studies uncertainty estimation through approximate Bayesian inference. MC Dropout uses stochastic dropout at inference time to approximate a Bayesian posterior predictive distribution [4], whereas Laplace-based methods approximate local posterior uncertainty around a trained solution [5, 6]. We chose to test a lightweight Laplace approximation because it is a computationally efficient way to capture local posterior uncertainty. It combines the simplicity of temperature scaling and the stochastic inference of MC Dropout. All three approaches provide practical alternatives to full Bayesian training and form the main methodological basis of our study.

Recent VLMs combine pretrained visual encoders with large language models and achieve strong performance on multimodal tasks such as VQA. BLIP-2 is a representative example of this paradigm, connecting frozen image encoders and frozen language models through a Querying Transformer [7]. Compared with standard image classification, calibration in multimodal systems is more challenging because uncertainty may arise from ambiguous visual evidence, linguistic variability in valid answers, and candidate-dependent inference. Calibration is also closely related to selective prediction, where a model may abstain when confidence is low [8]. Our work follows this practical direction by evaluating how standard post-hoc calibration methods affect both confidence quality and confidence-based abstention in BLIP-2 on VQA.

3 Method

We study post-hoc calibration for BLIP-2 on VQA without modifying the pretrained backbone. Given an image-question pair (x, q) , the model predicts an answer $\hat{y}$ with confidence $\hat{p}$. A model is well-calibrated when predictions made with confidence $\hat{p}$ are correct with probability p . In practice, however, VLMs are often overconfident, meaning that their predicted confidence exceeds their empirical accuracy.

In our setting, calibration is performed over a set of candidate answers rather than over a fixed classification head. For each sample, we construct a candidate set $\mathcal{C}(x, q) = \{c_1, \dots, c_K\}$ from the VQA annotations. BLIP-2 scores each candidate answer by conditioning on the image, question,

and candidate text, and produces a sequence score s_k . These scores are normalized with a softmax:

$$p(c_k \mid x, q, \mathcal{C}) = \frac{\exp(s_k)}{\sum_{j=1}^K \exp(s_j)}. \quad (1)$$

The predicted answer is the highest-probability candidate, and the confidence is

$$\hat{p} = \max_k p(c_k \mid x, q, \mathcal{C}). \quad (2)$$

Confidence is calculated by scoring each candidate answer with BLIP-2, using softmax to normalize those scores into a probability distribution, and taking the maximum probability as the model’s confidence in its predicted answer.

Our goal is to apply a post-hoc transformation to these candidate probabilities so that confidence better matches empirical correctness. We compare three post-hoc calibration methods. First, temperature scaling applies a scalar temperature $T > 0$ to the candidate scores:

$$p_T(c_k) = \frac{\exp(s_k/T)}{\sum_j \exp(s_j/T)}. \quad (3)$$

This simple, efficient method improves calibration by softening or sharpening output distribution. Second, MC Dropout estimates predictive uncertainty by enabling dropout during inference and averaging predictions over multiple stochastic forward passes:

$$\bar{p}(c_k) = \frac{1}{M} \sum_{m=1}^M p^{(m)}(c_k), \quad (4)$$

reducing overconfident predictions by making final confidence reflect variation across repeated runs.

Third, we test lightweight Laplace-style approximation. Instead of fitting full posterior over parameters, we perturb candidate scores with Gaussian noise and average resulting softmax probabilities:

$$\tilde{s}_k^{(m)} = s_k + \epsilon_k^{(m)}, \quad \epsilon_k^{(m)} \sim \mathcal{N}(0, \sigma^2), \quad (5)$$

$$\bar{p}_{\text{Lap}}(c_k) = \frac{1}{M} \sum_{m=1}^M \frac{\exp(\tilde{s}_k^{(m)})}{\sum_j \exp(\tilde{s}_j^{(m)})}. \quad (6)$$

This gives a cheap approximation to uncertainty, although it is less expressive than MC Dropout.

We also support abstention through confidence thresholding. If the predictive confidence is below a threshold τ , the model abstains instead of returning a normal answer. This allows us to examine whether better calibration also leads to more reliable abstention behavior.

4 Experimental Setup

4.1 Dataset

We use VQA v2 [9], a standard benchmark built on the Microsoft Common Objects in Context (COCO) dataset, a richly annotated visual dataset. VQA v2 supplements images from COCO with open-ended questions and ten human answers per question [9]. VQA v2 is well suited for studying both answer quality and confidence calibration because it combines visually rich scenes with answer ambiguity across different question types.

To keep inference manageable, we evaluate on a balanced subset drawn from the *yes/no*, *number*, and *other* answer categories. This is useful because calibration behavior can differ substantially across answer types: *yes/no* questions are often easier and more confident, whereas open-ended *other* questions typically exhibit greater ambiguity.

4.2 Model and Inference

Our base model is BLIP-2, which we treat as a frozen pretrained vision-language model [7]. We do not retrain the backbone. Instead, calibration is applied post hoc to the candidate-answer distribution produced at inference time.

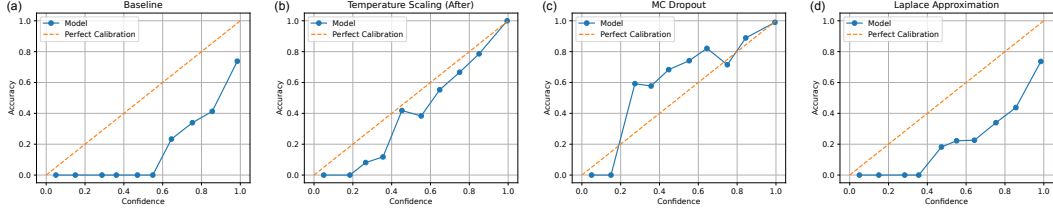


Figure 1: Reliability diagrams for the baseline model and post-hoc calibration methods. The ideal diagonal indicates perfect calibration. (a) Baseline, (b) Temperature scaling, (c) MC Dropout, and (d) Laplace approximation.

For each image-question pair, answers are scored against a candidate set derived from the human annotations. This avoids full free-form generation during evaluation and provides a practical way to compute candidate probabilities and confidence values in a consistent manner across methods.

4.3 Metrics

We evaluate model performance using accuracy, ECE, NLL, and reliability diagrams. Firstly, **accuracy** measures if the predicted answer matches the reference answer used for evaluation. **ECE** shows it by measuring the discrepancy between confidence and empirical accuracy across confidence bins:

$$\text{ECE} = \sum_{b=1}^B \frac{|I_b|}{n} |\text{acc}(I_b) - \text{conf}(I_b)|, \quad (7)$$

where I_b is the set of samples in bin b . **NLL** measures the negative log-probability assigned to the reference answer:

$$\text{NLL} = -\frac{1}{n} \sum_{i=1}^n \log p(y_i | x_i, q_i, \mathcal{C}_i), \quad (8)$$

where y_i denotes the reference answer for the i -th sample. Finally, **reliability diagrams** compare average confidence with empirical accuracy across bin to visualize calibration.

4.4 Implementation Details

Temperature scaling uses a held-out validation split to select the temperature parameter and is then evaluated on the test split. In our implementation, the temperature is selected by grid search over values from 0.1 to 5.0. MC Dropout performs repeated stochastic forward passes and averages the resulting candidate probabilities, while the lightweight Laplace-style method averages predictions obtained from Gaussian-perturbed score vectors. We additionally apply confidence thresholding to support abstention when the model is insufficiently certain.

5 Results

Table 1 summarizes the quantitative comparison of the four methods on our evaluation subset. Several observations follow from these results. First, the uncalibrated BLIP-2 baseline is clearly overconfident, with a high ECE of 0.321 and an NLL of 1.865. This confirms that raw confidence scores are poorly aligned with empirical correctness in our VQA setting. Second, temperature scaling provides the strongest calibration improvement. It reduces ECE from 0.321 to 0.097 and also achieves the best NLL. Although its accuracy is slightly lower than the baseline, it offers the most consistent improvement in confidence quality at almost no additional inference cost. Third, MC Dropout achieves the highest reported accuracy and also improves calibration relative to the baseline, reducing ECE to 0.112. However, in the final 500-sample setting, its calibration performance is slightly weaker than temperature scaling in both ECE and NLL. This suggests that MC Dropout provides a different trade-off: it yields strong answer quality and meaningful calibration gains, but does not produce the best overall calibration metrics in our final evaluation. Finally, the lightweight Laplace

Table 1: Comparison of post-hoc calibration methods. Higher is better for accuracy, while lower is better for ECE and NLL.

Method	Accuracy	ECE	NLL
Baseline	0.558	0.321	1.865
Temperature scaling	0.550	0.097	0.884
MC Dropout	0.815	0.112	1.013
Laplace approximation	0.578	0.301	1.864

approximation offers only limited benefit. Its ECE and NLL remain very close to those of the baseline, indicating that this simple approximation is not sufficient to capture uncertainty effectively in our multimodal setting.

The reliability diagrams in Figure 1 provide a complementary qualitative view of calibration. Panel (a) shows that the baseline model is strongly overconfident, with the empirical curve lying above the diagonal across most confidence bins. Temperature scaling (b) substantially improves calibration, bringing predictions closer to the diagonal, particularly in mid- to high-confidence regions. MC Dropout (c) further improves alignment overall, but exhibits slight underconfidence in several regions, where the curve falls below the diagonal. By contrast, the Laplace approximation (d) provides only modest improvement over the baseline, with persistent deviations from perfect calibration.

6 Discussion

The three post-hoc methods show different trade-offs. Temperature scaling provides the strongest calibration improvement in our final setting, achieving the lowest ECE and NLL with almost no additional inference cost. MC Dropout yields the highest reported accuracy and improves calibration relative to the baseline, but is slightly weaker than temperature scaling on the final calibration metrics and requires multiple forward passes. As reflected in Figure 1, MC Dropout can also exhibit mild underconfidence in some regions, where predicted confidence falls below empirical accuracy. The lightweight Laplace approximation provides only limited benefit, suggesting that simple Gaussian score perturbations are insufficient to capture uncertainty in this setting.

These results also highlight that calibration in VQA remains challenging even after applying post-hoc methods. As shown in Figure 1, none of the methods achieves perfect alignment with the diagonal across all confidence levels. In particular, deviations persist in lower-confidence regions and vary across methods, indicating that uncertainty in multimodal reasoning is not fully captured by simple post-hoc transformations. Uncertainty comes from the model and from ambiguous visual evidence and variation in acceptable answers. Thus, methods that transfer well from standard classification may still behave differently in multimodal reasoning. In practice, temperature scaling is the best low-cost default, while MC Dropout remains useful when additional computation is acceptable.

7 Conclusion

We studied confidence calibration in vision-language models through a post-hoc comparison of temperature scaling, MC Dropout, and a lightweight Laplace approximation on BLIP-2 for VQA. Our results show that the uncalibrated baseline is substantially overconfident, and that post-hoc calibration can improve reliability without retraining the backbone. Temperature scaling provides the strongest calibration improvement at very low cost, while MC Dropout achieves the highest accuracy and improves calibration overall, albeit with some residual miscalibration across confidence levels. The lightweight Laplace approximation offers only limited benefit. These findings should be interpreted with caution, as our evaluation is limited to VQA v2 and a single backbone, uses a candidate-based answer setup rather than full free-form generation, and includes minor implementation differences in scoring across methods. Overall, our results suggest that post-hoc calibration is a practical approach for improving the reliability of pretrained VLMs and highlight the need for future work to explore more expressive uncertainty estimation methods and broader calibration across diverse VLM architectures and multimodal tasks.

References

- [1] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C. Lawrence Zitnick, and Devi Parikh. Vqa: Visual question answering. *International Journal of Computer Vision*, 2017.
- [2] Xuesong Zhang, Jun He, Jia Zhao, Zhenzhen Hu, Xun Yang, Jia Li, and Richang Hong. Exploring and exploiting model uncertainty for robust visual question answering. *Nature*, 2024.
- [3] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2017.
- [4] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning*, 2016.
- [5] Agustinus Kristiadi, Matthias Hein, and Philipp Hennig. Being bayesian, even just a bit, fixes overconfidence in relu networks. *International Conference on Learning Representations*, 2020.
- [6] Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace redux – effortless bayesian deep learning. *Advances in Neural Information Processing Systems*, 2021.
- [7] Junnan Li, Dongxu Li, Silvio Savarese, and Steven C. H. Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning*, 2023.
- [8] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.
- [9] Yash Goyal, Tanmay Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the vqa dataset more robust to language priors. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017.

GenAI Disclosure

We used ChatGPT for some grammar, citation, and LaTeX editing.